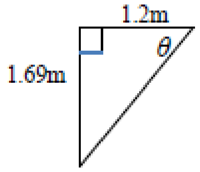
Q1.

(a)



$\theta = \tan^{-1}\dfrac{1.69}{1.2} = 54.623°$

AWRT 55°

B1

$u\cos 60° = v\cos 54.623°$

$v = 0.864u$

M1

$eu\sin 60° = v\sin 54.623°$

M1

$e = \dfrac{v\sin 54.623°}{\dfrac{v\cos 54.623°}{\cos 60°}\times\sin 60°}$

OE, dependent on both M1s

m1

$e = 0.813$ or 0.812

ISW

A1

5

(b) $I = 0.15u\sin 60° + 0.15v\sin 54.623°$

Single angle values needed for A1

M1A1

$= 0.15u\sin 60° + 0.15 \times \dfrac{u\cos 60°}{\cos 54.623°} \times \sin 54.623°$

m1

$= 0.236u$

AG (condone 0.2355 or negative result)

A1 **4**

(c) Attempt at considering motion parallel or perpendicular to AC

M1

$t = \frac{1.2}{u\cos 60^\circ}$

M1

$t = \frac{12}{5u}$ or $\frac{2.4}{u}$

OE, No ISW

A1 **3**

Alternative :

$CP = \frac{1.2}{\cos 54.623^\circ}$ (= 2.072703844 m)

(M1)

$t = \frac{\frac{1.2}{\cos 54.623^\circ}}{\frac{u\cos 60^\circ}{\cos 54.623^\circ}}$

(M1)

$= \frac{12}{5u}$ or $\frac{2.4}{u}$

(OE), No ISW

(A1) **(3)**

(d) Velocity (momentum) parallel to the cushion is unchanged, or,
Restitution only affects motion perpendicular to the cushion

Accept 'horizontal component of velocity is unchanged'

E1 **1**

[13]

Q2.

(a) Ball and plane are smooth $\Rightarrow$
Mutual reaction acts along the normal

to the plane at the point of impact $\Rightarrow$
No change in momentum parallel to plane

1 mark per implication

E2,1 **2**

(b) $\frac{3}{4} = \frac{v\sin\theta}{u\cos\theta}$ or $v\sin\theta = \frac{3}{4}u\cos\theta$

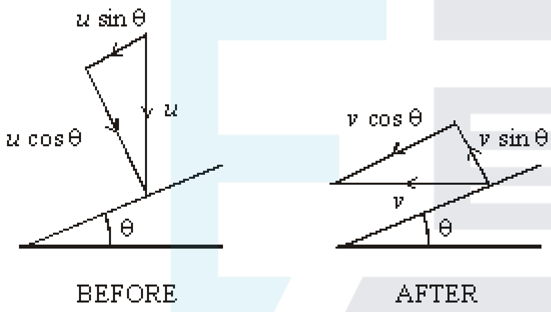
M1

$3u\cos\theta = 4v\sin\theta$

Answer given

A1 **2**

(c)



$v\cos\theta = u\sin\theta$

M1

$3u\cos\theta = 4u\tan\theta \ \sin\theta$

m1

$\frac{3}{4} = \tan^2\theta$

M1

$\theta = 40.9°$

Answer given

A1 **4**

(d) $I = mu\cos\theta - (mv\sin\theta)$

Impulse momentum

M1A1

$I = mu\cos\theta + mu\tan\theta\sin\theta$

m1

$I = \frac{mu}{\cos\theta}$

Elimination of v

A1F

$I = 1.32\, mu$

Answer given

A1

5

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Q3.

(a) (i) Velocities parallel to the wall

M1

2 u cos θ = 3u cos 60

A1

cos θ = $\frac{3}{4}$

A1

3

(ii) Restitution : 3u sin 60 . e = 2 u sin θ

M1A1

$e = \frac{2\sin\theta}{3\sin 60}$

= 0.509

A1

3

(iii) Angle is 60 + 41.4

M1

= 101.4

A1

2

(b) Impulse = change of momentum which is

M1

perpendicular to the wall
= $m.3u \sin 60 + m.\ 2u \sin \theta$

M1

= $3.92mu$

Needs + for second M1

A1

3

[11]

Q4.

(a) Ball leaves wall at an angle of 15°

B1

C of momentum along the wall:
$u \cos 30 = v \cos 15$

M1A1

$\therefore\ v = \frac{u \cos 30}{\cos 15}$

Restitution: $eu \sin 30 = v \sin 15$

M1

$\therefore\ e \sin 30 = \frac{\cos 30 \sin 15}{\cos 15}$

$e = \frac{\cos 30 \sin 15}{\sin 30 \cos 15}$

= 0.464

A1

5

(b) KE before impact = $\frac{1}{2} mu^2$

After impact = $\frac{1}{2} m \left(\frac{u \cos 30}{\cos 15}\right)^2$

= $0.4019\ mu^2$

B1

$\therefore$ Percentage loss in KE is $\frac{0.098}{0.5} \times 100$

Dependent on B1

M1

= 19.6%

Accept 19.5 or 19.7

A1

3

(c) Impulse = change in momentum perpendicular to the wall

M1

= mu sin30 + mv sin15

+

m1

= mu(0.5 + 0.232)

= 0.732mu or $\frac{1}{2}(1 + e)mu$

Accept mu(sin30 + cos30tan15)

A1

3

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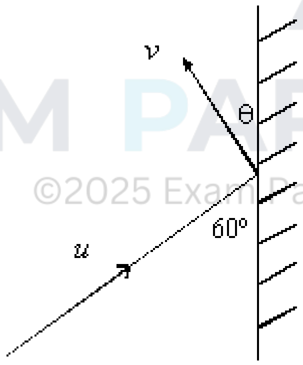
Q5.

(a)



Velocity // wall unaltered

M1

u cos 60 = v cos θ

A1

Velocity perp to wall

$e\ u$ sin 60 = v sin θ

M1A1

Dividing: tan $\theta = e$ tan 60

$= \frac{3}{4}\sqrt{3}$

$\therefore\ \theta = 52.4°$

A1

$\therefore$ Direction of motion is changed by 112.4°

A1

6

(b) Impulse is change in momentum perp to wall

M1

$= mu \sin 60 + mv \sin \theta$

M1

$= mu \sin 60\,(1 + e)$

A1

$= 0.3 \times 5 \times \frac{\sqrt{3}}{2} \times 1.75$

$= \frac{21}{16}\sqrt{3}$

A1

Time × Impulse = change in momentum

M1

$\therefore$ Impulse $= 20 \times \frac{21\sqrt{3}}{16}$

A1

= 45.5

A1

7

[13]

Examiner reports

Q1.

This question was answered fairly well by candidates. Part (b) of the question proved to be challenging for some candidates, where sign errors were often a source of mistake. For part (d), many candidates were unable to give a valid explanation of how they had used the assumption that the cushion was smooth in their answer. Some candidates made irrelevant comments about “line of centres”.

Q2.

Very few candidates answered part (a) of this question correctly. The main difficulty was a lack of understanding and appreciation of the effect of the smoothness of the ball and the plane on the mutual reaction force between them.

Part (b) of the question was answered correctly by almost all candidates.

A large number of candidates did not answer part (c) of the question, failing to utilise the given statement in part (a) viz. the component of the velocity of the ball parallel to the plane is not changed by the collision.

Some candidates could not answer part (d) of the question through not being able to eliminate v in their Impulse/Momentum equation by using the given statement in part (a).

Q3.

This question was well done. Many candidates gave the exact values in parts (a)(ii) and (b).

Part (a)(iii) was often incorrect but usually involved 41.4°. A few candidates did not find the velocities perpendicular to the wall in part (b), but most errors in this part were made by candidates ignoring directions perpendicular to the wall, finding $m.3u \sin 60 - m.2u \sin \theta$ rather than $m.3u \sin 60 + m.2u \sin \theta$.

Q4.

A number of candidates did not find 15º as the angle at which the particle left the wall. Most who did find this angle answered part (a) correctly. Part (b) was completed badly, with most attempting to consider only the motion perpendicular to the wall. Conversely in part (c), many candidates did not consider the velocities perpendicular to the wall, and of those who did, a number did not add the two components mu sin30 and mv sin15. A few gave the answer to part (c) as $\frac{1}{2}(1+e)mu$ and a small number of candidates used the exact value of tan 15, ie: $2 - \sqrt{3}$, to obtain the correct exact answer of $(\sqrt{3} - 1)mu$.

Q5.

Most candidates readily used the fact that the velocity parallel to the wall was unaltered in the impact. They also considered velocities perpendicular to the wall although with less success. Occasionally the signs used in this restitution equation were incorrect. Thus, many candidates correctly found the angle of 52.4° being the angle with which the particle left the wall. However a number did not then find the required angle to be 112.4°. In part (b), most candidates did not use the fact that the impulse (acting perpendicular to the wall) is the change in momentum **perpendicular** to the wall. Most candidates correctly multiplied the change in momentum by 20 (being 1/time) to find the impulsive force.